Swirl Jet Diffuser VSR



Description
Swirl jet diffuser with adjustable and radial arranged blades.

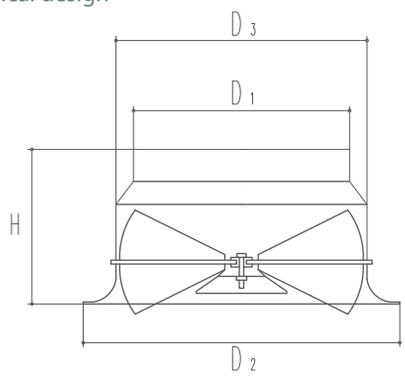
Application
Supply/ exhaust of the air.

Characteristics
Recommended for rooms over 3 m in height.
Circular and helical design promotes airflow swirling diffusion obtaining a high level of induction rate and reduces air stratification.

Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Aluminum diffuser electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design



D1	D2	D3	H
250	400	300	205
315	500	400	230
400	615	465	270
500	780	565	320
630	935	665	390

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Selection table VSR

Air flow (m^3/h)	Ø	250
	Ak [m^2]	0.050
250	Veff(m/s)	1.4
	NR [dB(A)]	<20
	X(m)	1.2
	△Pt [pa]	10.0
300	Veff(m/s)	1.7
	NR [dB(A)]	<20
	X(m)	1.5
	△Pt [pa]	15.0
350	Veff(m/s)	1.9
	NR [dB(A)]	22.0
	X(m)	1.6
	△Pt [pa]	20.0
400	Veff(m/s)	2.2
	NR [dB(A)]	26.0
	X(m)	1.8
	△Pt [pa]	28.0
500	Veff(m/s)	2.8
	NR [dB(A)]	33.0
	X(m)	2.8
	△Pt [pa]	45.0
600	Veff(m/s)	3.3
	NR [dB(A)]	38.0
	X(m)	3.6
	△Pt [pa]	70.0
700	Veff(m/s)	3.9
	NR [dB(A)]	44.0
	X(m)	4.6
	△Pt [pa]	91.0

Air flow (m^3/h)	Ø	315
	Ak [m^2]	0.078
500	Veff(m/s)	1.8
	NR [dB(A)]	<20
	X(m)	1.6
	△Pt [pa]	17.0
600	Veff(m/s)	2.1
	NR [dB(A)]	24.0
	X(m)	2.0
	△Pt [pa]	23.0
700	Veff(m/s)	2.5
	NR [dB(A)]	27.0
	X(m)	2.6
	△Pt [pa]	31.0
800	Veff(m/s)	2.8
	NR [dB(A)]	33.0
	X(m)	3.0
	△Pt [pa]	41.0
900	Veff(m/s)	3.2
	NR [dB(A)]	38.0
	X(m)	3.8
	△Pt [pa]	51.0
1000	Veff(m/s)	3.6
	NR [dB(A)]	43.0
	X(m)	4.2
	△Pt [pa]	69.0
1100	Veff(m/s)	3.9
	NR [dB(A)]	46.0
	X(m)	5.0
	△Pt [pa]	77.0

Air flow (m^3/h)	Ø	400
	Ak [m^2]	0.125
900	Veff(m/s)	2
	NR [dB(A)]	<20
	X(m)	1.8
	△Pt [pa]	13.0
1000	Veff(m/s)	2.2
	NR [dB(A)]	23.0
	X(m)	2.1
	△Pt [pa]	17.0
1100	Veff(m/s)	2.4
	NR [dB(A)]	26.0
	X(m)	2.5
	△Pt [pa]	21.0
1200	Veff(m/s)	2.7
	NR [dB(A)]	28.0
	X(m)	3.0
	△Pt [pa]	25.0
1400	Veff(m/s)	3.1
	NR [dB(A)]	32.0
	X(m)	3.8
	△Pt [pa]	33.0
1500	Veff(m/s)	3.3
	NR [dB(A)]	35.0
	X(m)	4.0
	△Pt [pa]	38.0
1600	Veff(m/s)	3.6
	NR [dB(A)]	38.0
	X(m)	4.5
	△Pt [pa]	44.0

Selection table VSR

Air flow (m^3/h)	Ø / Ak [m^2]	500 / 0.195
1200	Veff(m/s)	1.7
	NR [dB(A)]	30.0
	X(m)	1.8
	△Pt [pa]	17.0
1400	Veff(m/s)	2.0
	NR [dB(A)]	33.0
	X(m)	2.5
	△Pt [pa]	22.0
1600	Veff(m/s)	2.3
	NR [dB(A)]	36.0
	X(m)	3.0
	△Pt [pa]	28.0
1800	Veff(m/s)	2.6
	NR [dB(A)]	38.0
	X(m)	3.5
	△Pt [pa]	35.0
2000	Veff(m/s)	2.8
	NR [dB(A)]	41.5
	X(m)	4.0
	△Pt [pa]	42.0
2250	Veff(m/s)	3.2
	NR [dB(A)]	44.0
	X(m)	4.5
	△Pt [pa]	51.0
2500	Veff(m/s)	3.6
	NR [dB(A)]	47.0
	X(m)	5.2
	△Pt [pa]	70.0

Air flow (m^3/h)	Ø / Ak [m^2]	630 / 0.310
2100	Veff(m/s)	1.9
	NR [dB(A)]	25.0
	X(m)	3.5
	△Pt [pa]	18.0
2500	Veff(m/s)	2.2
	NR [dB(A)]	28.0
	X(m)	3.8
	△Pt [pa]	31.0
3000	Veff(m/s)	2.7
	NR [dB(A)]	35.0
	X(m)	4.8
	△Pt [pa]	38.0
3500	Veff(m/s)	3.1
	NR [dB(A)]	40.0
	X(m)	5.0
	△Pt [pa]	53.0
4000	Veff(m/s)	3.6
	NR [dB(A)]	44.0
	X(m)	5.5
	△Pt [pa]	72.0
4500	Veff(m/s)	4.0
	NR [dB(A)]	47.0
	X(m)	6.0
	△Pt [pa]	86.0
5000	Veff(m/s)	4.5
	NR [dB(A)]	50.0
	X(m)	7.0
	△Pt [pa]	110.0

Note:

The values from selection table are calculated for 45° blades inclination and 5 degrees temperature deviation.

Annotation

* Ak [m^2] - Free surface
* Veff[m/s] - Air velocity inside diffuser
* ΔPt [pa] - Pressure loss
* NR [dB(A)] - Noise level without the room attenuation
* X [m] - Air jet length

Order codes

Type	+	ADP IZ/N	/	Ø	/	0
VSR		Insulated/ Non-insulated adapter		250 315 400 500 630		**0** Painted white RAL **R** Other RAL paints